

# Deliverable 3: Test Cases Specification

Project: Secure Local Encrypted Password Vault Simulator | Pass Rate: 34/34 (100%)

## 1. Test Cases Coverage Table

The automated test suite covers normal operations, invalid inputs, boundary checks, duplicate rejections, tampering tests, rate-limiting lockout, and plaintext inspection:

| ID    | Category  | Test Scenario        | Input                 | Expected Outcome           | Status |
|-------|-----------|----------------------|-----------------------|----------------------------|--------|
| TC-01 | Normal    | Create Vault         | MasterPassword123!    | Vault created on disk      | PASS   |
| TC-02 | Invalid   | Password Mismatch    | Pass1 != Pass2        | AuthenticationError raised | PASS   |
| TC-03 | Boundary  | Short Password       | 'short' (<8 char)     | Rejection error message    | PASS   |
| TC-04 | Normal    | Unlock Vault         | Valid Password        | Vault unlocked             | PASS   |
| TC-05 | Invalid   | Wrong Password       | Incorrect password    | Authentication failed      | PASS   |
| TC-06 | Security  | Rate Limit Lockout   | 5 failed logins       | RateLimitExceededError     | PASS   |
| TC-07 | Normal    | Add Credential       | github/user/pass      | Stored encrypted           | PASS   |
| TC-08 | Normal    | Get Credential       | GET github            | Masked: *****              | PASS   |
| TC-09 | Normal    | Reveal Password      | REVEAL github + y     | Plaintext secret shown     | PASS   |
| TC-10 | Duplicate | Duplicate Service    | Add existing svc      | DuplicateServiceError      | PASS   |
| TC-11 | Invalid   | Unknown Service      | GET facebook          | ServiceNotFoundError       | PASS   |
| TC-12 | Normal    | Update Account       | UPDATE github         | Credential updated         | PASS   |
| TC-13 | Normal    | Delete Account       | DELETE github         | Credential deleted         | PASS   |
| TC-14 | Normal    | List Services        | LIST                  | Names only, no secrets     | PASS   |
| TC-15 | Normal    | Search Keyword       | SEARCH git            | Matched records list       | PASS   |
| TC-16 | Security  | Locked Access Block  | GET when locked       | VaultLockedError           | PASS   |
| TC-17 | Security  | Key Rotation         | New Master Pass       | Re-encrypted under new key | PASS   |
| TC-18 | Security  | Ciphertext Tampering | Bit-flip in file      | IntegrityError raised      | PASS   |
| TC-19 | Security  | Salt Tampering       | Tampered salt in JSON | Decryption rejected        | PASS   |
| TC-20 | Security  | Plaintext Inspection | Read raw disk file    | ZERO plaintext found       | PASS   |

## 2. Automated Test Execution Summary

```
$ pytest -v tests/
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0
collected 34 items

tests/test_auth.py (6 tests) ..... PASSED [ 17%]
tests/test_crypto.py (8 tests) ..... PASSED [ 41%]
tests/test_e2e.py (1 test) ..... PASSED [ 44%]
tests/test_security.py (4 tests) ..... PASSED [ 55%]
tests/test_storage.py (5 tests) ..... PASSED [ 70%]
tests/test_vault.py (10 tests) ..... PASSED [100%]

===== 34 passed in 5.19s =====
```